Learning Languages with Machines: Artificial Intelligence, Conversational Agents and Open Resources in Contemporary Foreign-Language Study

A critical review and an integrative framework

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Abstract

Purpose. Digital technology is no longer an optional supplement to foreign-language study; it is the environment in which much of that study now happens. This article asks a narrower and more answerable question than "do digital tools work": under what conditions do contemporary technologies — generative artificial intelligence (AI), AI-based conversational agents, automated feedback systems and open educational resources — strengthen second-language (L2) learning, and under what conditions do they merely create the impression of it? **Approach.** The article synthesises international research from applied linguistics, computer-assisted language learning (CALL) and the learning sciences into a single analytical framework. Three technological layers (input and resources, interaction, feedback and assessment) are linked to four pedagogical mechanisms (adaptive comprehensible input, negotiated interaction and pushed output, contingent feedback, and autonomy with affective safety) and four outcome families (linguistic, interactional, affective, metacognitive). **Findings.** Evidence is strongest where technology supplies something genuinely scarce — authentic input at the right level, immediate diagnosis, unlimited low-stakes practice — and weakest where it merely re-delivers conventional instruction through a new medium. Reported effect sizes are substantial (for example, $d = 0.95$ for corpus-based data-driven learning and $g = 0.87–0.99$ for captioned video) but are not directly comparable across designs. The most consequential risks are not technical failures but systemic ones: hallucinated content, false precision in automated scoring, learned dependence, accent and dialect bias in speech recognition, and the quiet redefinition of language ability as the ability to perform inside a particular application. **Implications.** The article proposes design principles for bot-mediated practice, a division of labour between automation and the teacher, quality gates for open resources, and a hypothetical 26-week roadmap illustrating how the pieces can be sequenced. It closes with a research agenda that prioritises delayed, transfer-oriented and equity-focused designs over short-term satisfaction surveys.

Keywords: artificial intelligence; generative AI; conversational agents; chatbots; computer-assisted language learning; open educational resources; second-language acquisition; feedback; learner autonomy; assessment validity

> **Note on terminology.** "AI" is used here in the ordinary sense of systems that perform tasks previously requiring human language or perception: automatic speech recognition (ASR), machine translation, natural-language processing, adaptive sequencing and large language models (LLMs). No claim is made about general intelligence. "Bots" refers to conversational agents of any architecture, from rule-based dialogue systems to LLM-based voice partners.

1. Introduction

For most of the twentieth century, the limiting resources in foreign-language education were the same three that had limited it in the nineteenth: access to competent speakers, access to varied authentic text, and access to individual attention. A learner in a mid-sized city who wished to study, say, Japanese or Portuguese depended on whatever local institution happened to offer, on printed materials that aged quickly, and on a teacher whose time was divided among thirty other students. The classroom solved the problem of structure — sequence, accountability, feedback — by accepting scarcity of input and interaction.

Digital technology has largely dissolved that particular scarcity, and in doing so has created a different problem. A motivated learner today can watch television from Osaka with same-language subtitles, query a corpus of two billion words, obtain an error correction within three seconds from a generative model, and hold a spoken conversation with a synthetic voice at two in the morning. None of these opportunities is hypothetical; all are free or nearly free. Yet the profession has not converged on an answer to the question that matters: which of these opportunities alter the underlying conditions of learning, and which merely make practice more pleasant?

The question is not academic. Investment in educational technology is routinely justified by claims of effectiveness that rest on supplier-run studies, short interventions and self-report satisfaction measures (Loewen et al., 2019; Williamson & Eynon, 2020). Meanwhile the rapid embedding of generative models into commercial products has moved the debate from "whether" to "how", often without an accompanying account of what the tools can be trusted to do. Reviews of the first wave of research on generative AI in language education report a heavy concentration on writing, a scarcity of quantitative outcome studies, and a frequent confusion between what learners *like* and what learners *learn* (Lo et al., 2024). Reviews of AI in education more broadly have reached similar conclusions for two decades: impressive technical capability, thin evidence of durable learning gains, and a persistent neglect of the pedagogical and institutional conditions under which tools are used (Zawacki-Richter et al., 2019; Holmes & Tuomi, 2022).

This article takes those findings as a starting point rather than a verdict. Its argument is that the effects of contemporary technology on language learning are neither intrinsic to the tools nor illusory. They are conditional: they depend on whether a given tool is deployed in a way that activates a known mechanism of second-language development. That claim is old — it is essentially the argument that media do not teach, methods do (Clark, 1983) — but it has become harder to sustain rhetorically now that the "medium" can generate content, adopt a persona and answer back. The task of this article is to restore the distinction.

Three questions organise the discussion:

1. **Descriptive.** What do contemporary technologies actually do, described in terms of the learning mechanisms they can plausibly activate (Section 3)?
2. **Evaluative.** What does the empirical record show about their effects, and how confident should we be in it (Sections 4–8)?
3. **Prescriptive.** How should learners, teachers and institutions sequence and govern these tools so that their use produces durable ability rather than the performance of ability (Sections 9–11)?

The contribution is integrative rather than empirical. The article offers (a) a framework that connects technological layers to pedagogical mechanisms and outcomes (Figure 1); (b) a taxonomy of AI conversational agents by pedagogical role (Figure 3) together with design principles; (c) a description of the AI-in-the-loop architecture and its conditions of validity (Figure 2); (d) a map of the open-resource ecosystem with quality gates (Figure 5); (e) a critical reading of the effect-size literature (Figure 6); and (f) a hypothetical, explicitly illustrative 26-week roadmap (Figure 7) that shows how the components can be sequenced. Throughout, the aim is to state plainly what is known, what is plausible, and what is currently asserted without adequate evidence.

The remainder of the article is structured as follows. Section 2 situates the present moment historically and theoretically. Section 3 introduces the framework. Sections 4 to 7 examine, in turn, generative AI, AI-based bots, automated feedback and open resources. Section 8 assesses the evidence base and its methodological limits. Section 9 presents the roadmap. Section 10 addresses risk, ethics and equity. Section 11 draws implications and sets out a research agenda. Section 12 concludes.

2. From CALL to AI-mediated language learning

2.1 Three generations, one recurring question

Computer-assisted language learning has passed through three broad generations, each defined less by its hardware than by the metaphor of the machine it encoded (Table 1). The first generation treated the computer as a tutor: drills, immediate right-or-wrong judgements, programmed instruction. The second treated it as a tool and then as a medium: word processors, concordancers, email, discussion boards, later the web and its communities. The third, still forming, treats the computer as a participant: an interlocutor that produces language, adopts roles and responds to meaning rather than to a predetermined answer key.

Each generation provoked the same objection, and each time the objection was partly right. Drill-and-practice software of the 1980s produced gains on the forms it drilled and little else. Early computer-mediated communication produced fluency and confidence but often at the cost of accuracy, since no one corrected anything. The lesson of that history is not that one metaphor is superior but that each activates a different mechanism, and that a mechanism requires specific conditions to operate.

Table 1. Three generations of technology in language learning, by dominant metaphor, typical mechanism and characteristic weakness.

Generation	Dominant metaphor	Representative tools	Mechanism most strongly activated	Characteristic weakness
1960s–1990s: CALL	Machine as tutor	Language labs, drill-and-practice software, early intelligent tutoring systems	Repeated retrieval; immediate knowledge-of-result feedback	Decontextualised forms; transfer to communication rarely tested
1990s–2010s: CMC and MALL	Machine as medium and tool	Email projects, discussion boards, wikis, mobile apps, concordancers	Negotiated interaction; authentic input; learner autonomy	Accuracy drift; time-on-task measured instead of learning; novelty effects
2015s–present: AI-mediated	Machine as participant	ASR-based pronunciation tools, adaptive platforms, conversational agents, LLMs	Contingent feedback; pushed output; adaptive input; simulated interaction	Hallucination; false precision; dependence; bias; opacity of the learner model

2.2 What is genuinely new

Not everything labelled “AI” is new, and not everything new is AI. Speech recognition, machine translation and intelligent tutoring have research histories measured in decades. Four discontinuities nevertheless justify treating the present period as distinct.

First, **generative fluency**. LLMs produce extended, coherent, register-appropriate text and speech on demand, at any specified level of difficulty, on any topic, in dozens of languages. For language pedagogy this is a change of kind rather than degree: the binding constraint on practice material has always been the cost of production, and that cost has fallen close to zero.

Second, **responsiveness to meaning**. Rule-based dialogue systems required predictable learner moves and failed on anything unanticipated. LLM-based agents tolerate unpredictable input, negotiate topics and, crucially, can reformulate learner output rather than merely judge it. This makes possible something closer to the interactional feedback that classroom research has long identified as facilitative (Long, 1996; Lyster & Ranta, 1997).

Third, **multimodality**. Contemporary systems accept and produce text, speech, images and video. A learner can photograph a sign and ask what it says, or describe a picture aloud and receive correction on both content and pronunciation. Multimodal input is not a luxury: dual coding of verbal and visual information is a well-established facilitator of comprehension and retention (Mayer, 2009).

Fourth, **programmability with natural language**. Teachers can now construct a task, a rubric or a level-adjusted text by describing it. This shifts the locus of materials design from publishers to practitioners, and from code to prose. It also means that the quality of a learning activity increasingly depends on the quality of a prompt — a skill that is not yet part of most teacher-education curricula (Godwin-Jones, 2024).

2.3 Theoretical anchors

Technology does not generate its own theory of learning. The mechanisms that contemporary tools can activate are the ones second-language acquisition (SLA) research has described for forty years, and it is against them that the tools should be judged.

- **Input.** Learners develop by processing comprehensible input slightly beyond their current level (Krashen, 1985), and comprehension is supported when input is multimodal, repeated and interactionally modified (Long, 1996). Adaptive systems and captioned video are, in these terms, input-delivery technologies.
- **Noticing and output.** Development requires attention to form (Schmidt, 1990) and opportunities to produce language under communicative pressure (Swain, 2005). A bot that only supplies input, however pleasant, does not satisfy the output condition; a bot that pushes the learner to reformulate does.
- **Interaction and feedback.** Negotiation of meaning and corrective feedback — recasts, elicitations, metalinguistic explanation — create the conditions under which learners notice gaps (Long, 1996; Lyster & Ranta, 1997). Automation is most defensible precisely where it increases the *frequency* and *timeliness* of such feedback.
- **Mediation and autonomy.** Learning is mediated by tools and by more capable others within a zone of proximal development (Vygotsky, 1978). Technology-mediated autonomy has been a central concern of CALL since its institutionalisation (Reinders & White, 2016): the question is not whether learners can act alone, but whether they can regulate their own activity with the resources available.
- **Motivation and affect.** Persistence depends on competence, autonomy and relatedness (Ryan & Deci, 2000), and on the learner's belief that success is attainable (Bandura, 1997). Anxiety, a well-documented inhibitor of oral production, is particularly relevant to bot-mediated speaking practice.
- **Cognitive offloading.** Humans readily delegate cognitive work to external aids and thereby reduce the effort that produces durable learning (Risko & Gilbert, 2016). This is the mechanism behind the most under-reported risk of AI assistance in language study.

3. A framework: three layers, four mechanisms, four outcomes

To compare tools that differ in architecture but not in purpose, it is useful to separate three questions: what the technology supplies, what pedagogical process it initiates, and what change it can plausibly produce. Figure 1 organises these questions as three vertical layers connected by arrows, with a band of moderators below.

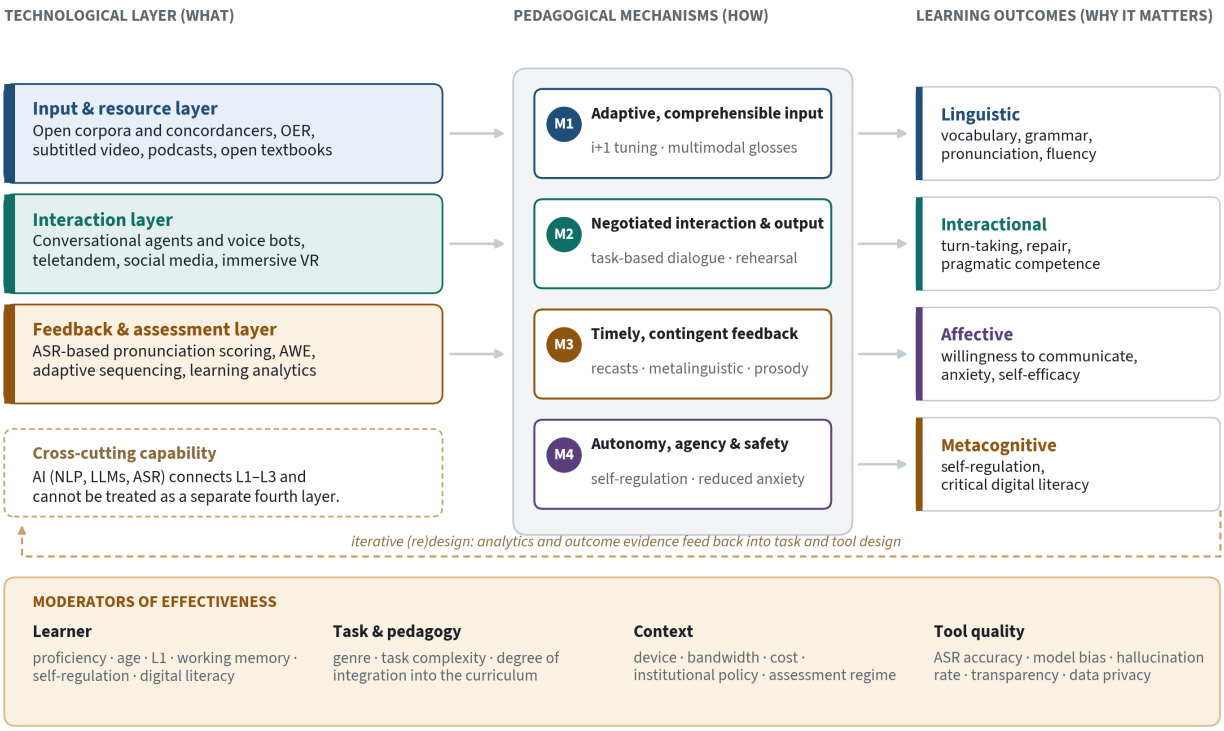


Figure 1. An integrative framework. Technological layers supply pedagogical mechanisms, which in turn act on four outcome families. Artificial intelligence is treated as a cross-cutting capability rather than as a fourth layer. The band below lists moderators that condition every pathway; the dashed line indicates that outcome evidence feeds back into task and tool design.

Layer 1 — input and resources. Open corpora and concordancers, open textbooks, subtitled video, podcasts and open courses supply language and information about language. Their pedagogical value depends on level-appropriateness, authenticity and the presence of scaffolds (Section 7).

Layer 2 — interaction. Conversational agents, voice assistants, teletandem partnerships, social media and immersive environments create occasions to use the language with a partner, human or synthetic. Their value depends on whether the interaction contains negotiation, repair and opportunities to produce extended output (Sections 5 and 7).

Layer 3 — feedback and assessment. ASR-based pronunciation scoring, automated writing evaluation, adaptive item selection and learning analytics diagnose performance and report on it. Their value depends on validity: whether the score means what it is taken to mean (Section 6).

Three features of the framework deserve emphasis. First, the layers are not a hierarchy: a learner may work productively at any layer, and the same tool may act at more than one. Second, AI is drawn as a cross-cutting capability precisely because it is not a separate pedagogical category; it changes the economics of the three layers rather than adding a fourth. Third, the framework is de-

liberately mechanistic. It allows the claim "this tool works" to be decomposed into the more answerable claims "this tool activates this mechanism, for this learner, under these conditions" — a decomposition that most marketing claims and a good deal of research fails to make.

Four outcome families close the right-hand column. **Linguistic** outcomes include vocabulary, grammar, pronunciation and fluency. **Interactional** outcomes include turn-taking, repair and pragmatic appropriateness — the ability to *do things* with the language. **Affective** outcomes include willingness to communicate, anxiety and self-efficacy. **Metacognitive** outcomes include self-regulation and critical digital literacy. The last two are frequently omitted from evaluations, which is a serious omission: an intervention that raises measured proficiency while lowering willingness to speak has not obviously succeeded.

The moderators in the lower band are not decoration. The same chatbot produces different effects for a B2 learner rehearsing a job interview and an A1 learner who cannot yet parse its output; the same corpus task succeeds in a seminar room with a teacher and fails in a commuter train. Reviews of technology-mediated learning consistently find that design and context variables account for more variance than tool identity (Plonsky & Ziegler, 2016).

4. Generative artificial intelligence

4.1 Four modes of use

Generative AI enters language study through four distinguishable doors, which are often conflated in both research and marketing. Separating them matters because each activates a different mechanism and carries a different failure profile.

As tutor. The model explains grammar, defines words in context, contrasts the learner's choices with alternatives, and answers the question the learner is embarrassed to ask in class. This is the mode with the longest research pedigree in intelligent tutoring and the clearest mechanism: metalinguistic feedback delivered on demand, at the moment of need (Barrot, 2024). Its weakness is that explanations can be confidently wrong, and that learners cannot easily detect wrongness in a language they do not yet control.

As generator of material. The model produces graded texts, exercises, dialogues, comprehension questions, model answers and tests, at a specified level, on a specified topic. Cost of production collapses, which makes possible the differentiation that teachers have always wanted and rarely had time to produce (Kohnke et al., 2023). Its weakness is level control: models systematically overestimate the simplicity of their own output unless constrained by explicit criteria such as sentence length, frequency bands or a target structure list.

As writing partner. The model drafts, rewrites, summarises and corrects. This is by far the most studied application, and the one with the most polarised evidence: gains in text quality and revision behaviour coexist with evidence of reduced ownership, reduced complexity in learners' own subsequent writing, and serious validity problems for take-home assessment (Barrot, 2023; Lo et al., 2024). The pedagogically defensible version is not "write with AI" but "write, then interrogate the AI's revision and justify or reject each change".

As interlocutor. The model plays a role — a hotel receptionist, a sceptical examiner, a fellow student who disagrees — and sustains it across turns. This is the mode with the greatest unrealised potential, because it addresses the scarcest resource of all: a patient partner available at any hour. It is treated separately in Section 5.

Table 2. Four modes of generative-AI use, with the mechanism each can activate and the principal risk attached to it.

Mode	Typical learner action	Mechanism activated	Evidence status	Principal risk
Tutor	Ask "why is this wrong?"	Contingent metalinguistic feedback	Moderate; mostly short interventions	Confident inaccuracy; learner cannot audit
Generator	Request a B1 text on X with a glossary	Adaptive comprehensible input	Emerging; little controlled evidence	Level drift; unverified facts and collocations
Writing partner	Draft, receive a rewrite, compare	Noticing; revision; focus on form	Substantial but mixed; heavy concentration of studies	Reduced ownership; assessment invalidity
Interlocutor	Role-play a task by voice or text	Negotiated interaction; pushed output	Growing; small samples, short duration	Shallow praise; avoidance of repair; drift from role

4.2 Why "the model is the method" is the wrong assumption

A recurrent error in both practitioner and research discourse is to attribute learning effects to the model rather than to the interaction design around it. The same underlying model produces radically different learning experiences depending on four decisions that are entirely pedagogical:

1. **The task.** A model asked to "chat in Spanish" will produce pleasant, undemanding conversation. A model asked to play a receptionist who has lost the learner's reservation, and who will not resolve the problem until the learner has requested, complained and negotiated a solution, produces a different event entirely.
2. **The constraints.** Specifying the target level, the register, the maximum number of new words per turn, and an explicit instruction not to accept interlanguage forms converts a chatbot into a scaffolded interlocutor.
3. **The feedback regime.** Left to itself, a model tends to respond to meaning and ignore form — the same behaviour that makes human conversation fluent and makes it a poor language lesson. Deliberate instruction to recast one error per turn, or to withhold correction until a task is complete, changes what the learner can notice.
4. **The closure.** Sessions that end with the learner producing a summary, a written follow-up or a re-performance of the same task produce artefacts that can be reviewed, by the learner or the teacher. Sessions that end when the learner gets bored produce nothing.

4.3 The architecture of an AI-in-the-loop environment

Figure 2 sets out the components of a defensible AI-mediated learning environment. The learner supplies text, speech, images or media. An orchestration layer — a large language model combined with speech models — performs five functions: role and scenario design; retrieval of relevant evidence from corpora, syllabus documents, rubrics or glossaries; maintenance of a learner model; generation with revision; and guardrails. The outputs are pedagogical: feedback, interaction and content. Telemetry flows back into the learner model, and a set of conditions of validity constrains the whole.

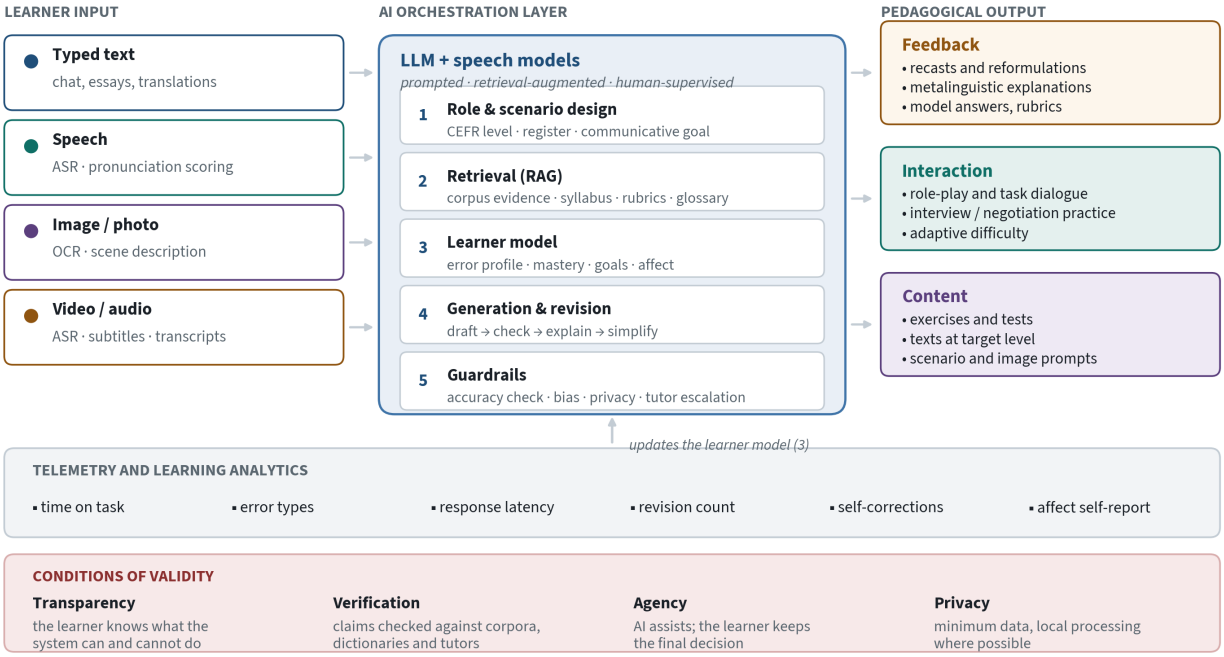


Figure 2. Architecture of an AI-in-the-loop language-learning environment. The orchestration layer converts learner input into feedback, interaction and content; telemetry updates the learner model; the conditions of validity in the lower band determine whether the system's outputs are pedagogically usable.

Two components deserve comment because they are usually missing in practice.

Retrieval. A model that answers from its parameters alone may produce plausible but unattested collocations, invented etymologies and fabricated grammatical rules. Retrieval-augmented generation — constraining output by reference to a corpus, a dictionary or a verified syllabus — converts an unreliable oracle into a checkable one. For language teaching this is not a technical refinement but a validity requirement: the learner is trying to acquire the *target* language, not the model's approximation of it.

Guardrails. The final box in the orchestration layer is not a filter but a division of labour. It should specify what the system will refuse to do, when it must escalate to a human, what it must never assert without evidence, and what data it may retain. The conditions of validity in Figure 2 — transparency, verification, agency and privacy — are properties of the *system*, not of the learner's goodwill, and they must be designed in.

4.4 What the evidence supports

The empirical literature on generative AI in language education is young, fast-moving and uneven. Systematic reviews of the first two years report three consistent patterns (Barrot, 2023, 2024; Kohnke et al., 2023; Lo et al., 2024): a strong concentration on writing; predominantly positive findings on immediate measures of quality and self-efficacy; and a scarcity of designs capable of supporting causal claims about durable learning. A review of empirical studies in ESL/EFL contexts concluded that most work examined learners' use of the tool rather than its effects on learning, and that quantitative outcome evidence remained limited (Lo et al., 2024).

Three findings are nonetheless robust enough to guide practice. First, generative feedback is *accepted* and *used* by learners, particularly when it is immediate and specific; engagement with feedback is not the bottleneck. Second, learners' self-efficacy — their belief that they can perform — rises after structured use of AI conversation features (Kittredge et al., 2025). Third, effects depend heavily on how the tool is integrated: studies in which AI feedback was followed by revision and re-submission report larger gains than studies in which the output was simply displayed. The pattern is consistent with the general finding in the feedback literature that what matters is uptake, not delivery. Independent efficacy studies of mobile applications point in the same direction: learners who complete beginning-level courses in a mainstream app reach reading and listening levels comparable to several semesters of university study (Jiang et al., 2021), while speaking and interaction remain the weakest outcomes (Loewen et al., 2019, 2020).

What the evidence does not support is equally important. There is no credible basis for claiming that generative AI substitutes for instruction, that it accelerates acquisition of a language from zero without human mediation, or that its scores on production tasks are equivalent to those of trained raters. These claims circulate widely; they are not findings.

5. AI-based conversational agents

5.1 A taxonomy by pedagogical role

"Chatbot" is a technological category that conceals a pedagogical one. A single model can be asked to behave as a drill partner, a conversational peer, a tutor, an examiner, a tutee or a content generator, and these roles differ more from each other than the underlying architectures do. Figure 3 organises agents by role, showing the mechanism each activates and the characteristic failure mode of each.

Three points follow from the taxonomy. First, **role determines evidence requirements**. A drill partner can be evaluated on retention of forms; a conversational peer must be evaluated on interactional and affective outcomes; an examiner must be evaluated on the validity of its scores. Treating these as one intervention is a category error that pervades the literature.

Second, **the tutor role is the most over-rated and the tutee role the most under-explored**. Tutor agents are popular because they are easy to build, but explanation is exactly the activity in which a plausible-sounding error is most damaging. Conversely, asking the learner to *teach* the agent — with the agent deliberately making the errors the learner typically makes — exploits the well-established benefits of learning by teaching and produces diagnostic evidence that a tutor agent does not.

Third, **locus of control matters**. Learner-directed agents support autonomy and exploration; system-directed agents support assessment and accountability. Most institutional deployments default to the latter because it produces data, which creates a subtle drift away from the conditions in which learners actually talk.

5.2 What the evidence shows

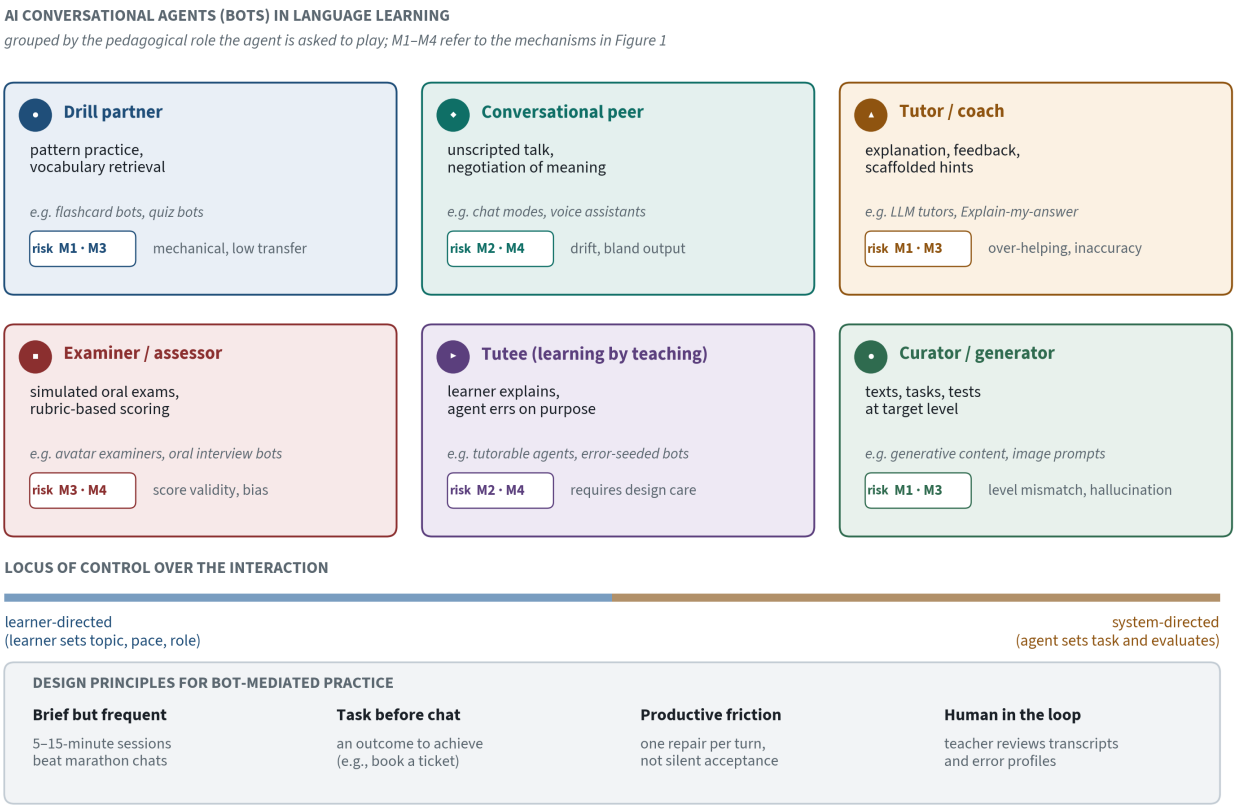


Figure 3. A taxonomy of AI conversational agents by pedagogical role, with the mechanisms they primarily activate (M1–M4 as in Figure 1) and their characteristic risks. The continuum below indicates who controls the interaction; the lower band summarises design principles.

Meta-analytic and systematic evidence on conversational agents in language learning is more favourable than the evidence on generative AI generally, partly because it rests on a longer tradition of dialogue-based CALL. A systematic review of AI-powered chatbots for EFL speaking practice identified consistent reported benefits for speaking practice, feedback and motivation, while noting short intervention durations and heterogeneous outcome measures (Du & Daniel, 2024). A controlled mixed-methods study with 60 IELTS candidates found significant gains in speaking performance after six weeks of bot-mediated practice (d = 0.60) together with a significant reduction in speaking anxiety (Ding & Yusof, 2025). A multilevel meta-analysis of 17 effectiveness studies (100 effect sizes) found a moderate overall effect of dialogue-based CALL on L2 development (d = 0.58), sustained on delayed post-tests (d = 0.56), and identified three moderators that map directly onto the design principles below: goal-oriented systems, systems that constrain the meaning of learner production, and the provision of corrective feedback were all associated with larger effects (Bibauw et al., 2022). In other words, the meta-analytic evidence supports the pedagogical design, not the chatbot as such.

The affective finding deserves particular attention. Foreign-language anxiety is one of the best-documented obstacles to oral development, and the social conditions that produce it — evaluation by a peer, fear of appearing foolish, the irrevocability of an utterance — are absent or attenuated in interaction with a machine. Learners consistently report lower anxiety and greater willingness to speak with agents than with humans (Ding & Yusof, 2025), and this is a genuine mechanism, not a measurement artefact. But it has a boundary condition: the anxiety that is avoided in practice is

precisely the anxiety that must eventually be managed in performance. A pedagogical sequence that never reintroduces social risk can produce a learner who is fluent with the machine and mute with the waiter.

5.3 Design principles for bot-mediated practice

The principles in the lower band of Figure 3 are derived from the mechanisms in Section 2.3 rather than from the technology, and they are the operational content of the "conditional effects" claim made in the introduction.

Brief but frequent. Sessions of five to fifteen minutes, repeated across days, are more consistent with what is known about distributed practice and retrieval than marathon conversations. Long sessions also drift: as the conversation grows, task pressure relaxes and the interaction becomes social chat.

Task before chat. The agent should be given an outcome to withhold until the learner achieves it. Booking a ticket, complaining about a defective product, persuading a landlord, passing an oral examination — each generates the communicative pressure that pushes output. Conversation without a task produces fluency in conversation about nothing in particular.

Productive friction. A well-designed agent repairs. It signals misunderstanding, requests clarification, reformulates, and declines to accept interlanguage forms that a competent speaker would not understand. An agent that praises everything trains the learner to produce output that is never checked against reality. This is the single most consequential design decision in bot-mediated practice, and the one most often sacrificed for user satisfaction.

Human in the loop. Transcripts and error profiles should be reviewable by a teacher, and periodically reviewed. This serves two purposes: it allows the teacher to detect systematic errors that the learner cannot see, and it creates accountability for the automated feedback.

Table 3. Design parameters for a task-based bot session, with the mechanism each parameter targets (Section 5.3).

Parameter	Recommended specification	Mechanism targeted
Duration	5–15 minutes per session; several per week	Distributed practice; sustained attention
Task	One transactional or interpersonal goal per session	Pushed output; communicative pressure
Level	Explicit CEFR target plus a maximum of two new items per turn	Comprehensible input
Error policy	One recast or clarification request per turn; no correction of errors that do not impede meaning	Noticing without disruption of fluency
Refusal policy	Agent does not accept an incomprehensible or inappropriate request	Negotiation of meaning
Closure	Learner summarises the outcome in writing or re-performs the task	Consolidation; evidence for review
Review	Transcript and error profile available to learner and teacher	Human oversight; self-regulation

5.4 Spoken interaction, pronunciation and immersive environments

The addition of speech recognition and synthesis turns the text agent into a speaking partner, and here the technical quality of the components becomes pedagogically consequential. Modern ASR handles read speech well and spontaneous non-native speech less well; accuracy varies by accent, by first language and by recording conditions (Koenecke et al., 2020). Pronunciation scoring, usually produced as a composite or per-phoneme score against a model of native speech, is reliably sensitive to gross segmental errors and much less reliable for prosody, connected speech and the intelligibility dimensions that actually matter in communication.

The instructional implication is not to abandon ASR but to use it where its validity is highest: minimal-pair discrimination, word and phrase stress, segmental accuracy in controlled production, and — most defensibly — as a *practice* device whose scores are treated as formative signals rather than as measurements. Reporting an ASR-derived score as a proficiency level is a validity error, discussed further in Section 6.

Immersive virtual environments add spatial presence and social copresence to the interaction, and systematic reviews report benefits for situated communication, reduced anxiety in some designs, and high equipment and design costs (Parmaxi, 2023). Their present significance for language education is less as a mass medium than as a laboratory for studying what presence contributes to interaction — a question that becomes more rather than less important as synthetic agents become more convincingly embodied.

6. Automated feedback and assessment

6.1 The loop

Feedback is where technology has the clearest mechanism and the greatest risk of false confidence. Figure 4 presents the loop as six stages, each of which may be automated, augmented or retained by the teacher: performance, diagnosis, feedback design, uptake, revision, and oversight.

Three observations follow from the figure. First, **the loop is only closed by uptake.** Most evaluations measure the quality of the feedback rather than its uptake, and a technically excellent correction that the learner ignores has no learning value. Technologies that force a subsequent action — a revision, a retry, a justification — close the loop; technologies that merely annotate do not.

Second, **automation is not equally reliable at all six stages.** Flagging a missing article, scoring a word against a pronunciation model and generating ten variants of an exercise are tasks that machines do reliably at scale. Judging whether a rhetorical move is appropriate for the audience, deciding that a particular error is worth teaching now, and interpreting a learner's silence as disengagement are not. The division of labour in the right-hand panel of Figure 4 is therefore not a compromise between humanists and technologists; it is a description of where each party's comparative advantage lies.

Third, **two failure modes must be designed against explicitly.** *False precision* occurs when a system reports a confident number that overstates what was measured — a "pronunciation score of 78" that reflects segmental accuracy in read speech and says nothing about intelligibility. *Learned dependence* occurs when the learner's successful performance is produced by the tool rather than by the

learner, and it is invisible in the app because the app observes the performance, not the competence behind it. Both failure modes are detectable: the first by validation studies against human raters, the second by transfer tasks performed without assistance.

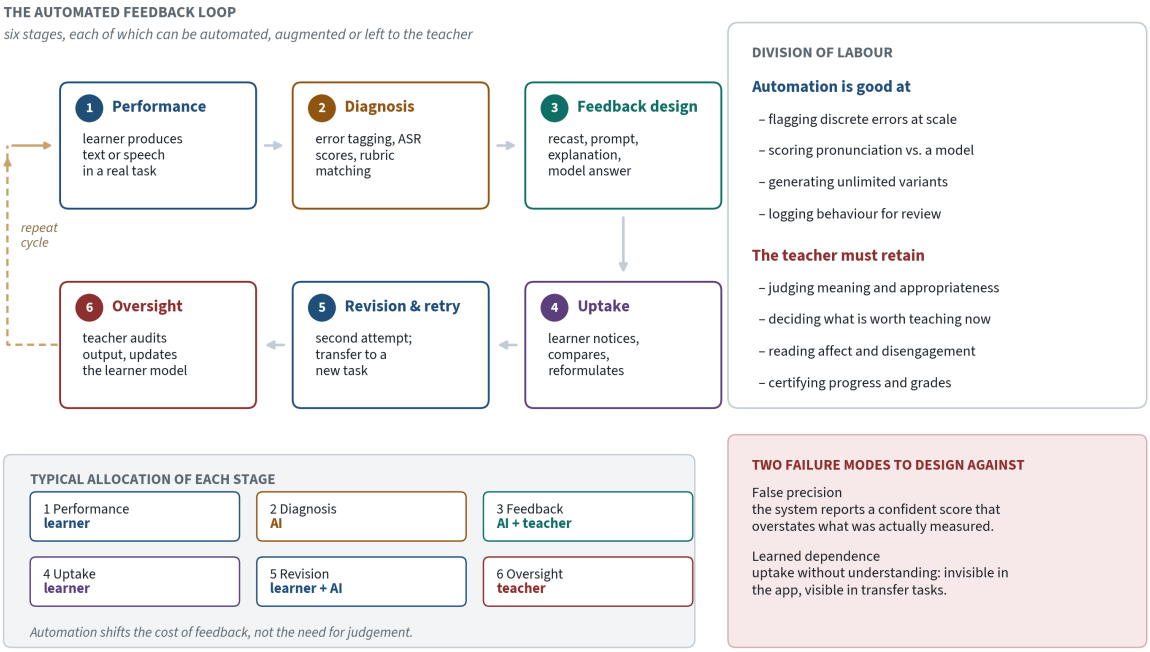


Figure 4. The automated feedback loop. Automation is most reliable at diagnosis and least reliable at judging meaning and appropriateness; the lower band summarises the division of labour and names the two failure modes that design must prevent.

6.2 Automated writing evaluation and the feedback that learners can use

Automated writing evaluation has a thirty-year history and a reasonably clear evidence base. Its strengths are consistency and speed: it applies the same criteria to every submission, provides feedback within seconds, and does not tire. Its weaknesses are equally clear. It tends to reward surface features that are easy to detect — length, lexical diversity, syntactic complexity, discourse connectives — and to under-detect problems of argumentation, register and rhetorical fit. Learners quickly learn to optimise for the detector, producing prose that satisfies the metric and communicates less.

The most useful configuration of AWE in language teaching is therefore neither fully automated nor fully manual but *sequenced*: automated feedback on the features it handles well (mechanics, agreement, lexical repetition, cohesion at sentence level), followed by peer or teacher feedback on meaning and appropriateness, followed by revision. Studies that embed automated feedback within such a sequence report better outcomes than studies that substitute it for human feedback, a pattern consistent with the broader finding that feedback effects depend on the learner's opportunity and obligation to act on it.

6.3 Validity, fairness and the limits of scoring

Assessment is where the commercial and pedagogical pressures on AI are strongest, and where the conceptual issues are clearest. Three deserve particular attention.

Construct validity. A test is valid insofar as it measures the construct it claims to measure. An automated speaking test that scores pronunciation, fluency and vocabulary but cannot evaluate interactional competence is not a test of speaking in any rich sense; it is a test of three measurable components of speaking. This is not a fatal objection, provided the score is reported for what it is. It becomes fatal when an institution treats the score as a certificate of communicative ability.

Fairness across accents and varieties. ASR and scoring models are trained on particular distributions of speech. Where those distributions under-represent a learner's accent, region or first language, error rates rise and scores fall for reasons unrelated to proficiency. The fairness problem is not incidental: it systematically disadvantages exactly the learners who most need reliable assessment, and it is compounded when the underlying model is opaque (Holmes & Tuomi, 2022).

Assessment security. Generative models have made take-home writing assessment unreliable as a measure of the learner's own ability. Institutions have responded with detection software, whose error rates are unacceptable, and with a return to in-person and process-based assessment. The more durable response is to assess what cannot be outsourced: live interaction, oral defence of written work, portfolio evidence with revision history, and tasks performed under observation.

7. Open resources and the curation problem

7.1 Six families of resources

The open-resource ecosystem is the least glamorous and most immediately useful part of the digital turn. Its components are well established, widely available and, in many cases, free. Figure 5 groups them into six families and pairs each with the pedagogical use it supports.

Corpora and datasets. Concordancers make it possible to ask questions that dictionaries answer badly: which preposition does this verb actually take, how frequent is this construction in academic prose versus conversation, what does this word collocate with. Data-driven learning — the use of corpus queries as a learning activity — has the strongest meta-analytic support of any technology in this review: a synthesis of 64 studies reported an average effect of $d = 0.95$ for between-group comparisons and $d = 1.50$ for pre/post designs (Boulton & Cobb, 2017). Learner corpora add a further affordance: comparing one's own errors with those of a comparable population normalises error and supports self-diagnosis.

Lexical resources. Community dictionaries, WordNets and semantic networks, and banks of subtitles and parallel texts provide definitions, examples, inflectional information and phraseology in context. Their weakness is variable quality control; their strength is coverage of precisely the low-frequency, domain-specific and idiomatic language that textbooks omit.

Open educational resources. Open textbooks, lesson repositories and open courses provide structure. The difficulty is not availability but discoverability and level alignment: the ecosystem contains an enormous quantity of material whose level is unlabelled, inconsistently labelled or labelled by a different framework than the one the learner uses.

Audiovisual media. Public-service broadcasting, podcasts and captioned video supply authentic language in quantity. The evidence on captions is among the most consistent in the field: a meta-analysis of 18 studies reported large effects for listening comprehension ($g = 0.99$) and vocabulary learning ($g = 0.87$), with larger effects for receptive than for productive tests (Montero Perez et al.,

2013). Captions work because they make input comprehensible through a second modality while keeping the learner in contact with the spoken signal — a genuine dual-coding advantage (Mayer, 2009).

Communities and exchange. Teletandem and virtual exchange pair learners across institutions for authentic interaction; forums and social media offer participation in real genres; volunteer tutoring offers low-cost speaking practice. The evidence base for virtual exchange is substantial and consistently reports gains in intercultural competence, interactional confidence and digital literacy, alongside persistent logistical fragility (O'Dowd, 2018). Social media use in language learning has been reviewed extensively; the recurring finding is that benefits depend on task design and on the learner's purpose rather than on the platform (Reinhardt, 2019).

Open tooling. Flashcard systems with spaced retrieval, subtitle and transcript tools, and locally run AI models complete the picture. The last of these is increasingly significant for privacy: a model running on the learner's own device can provide practice without transmitting audio or text to a third party.

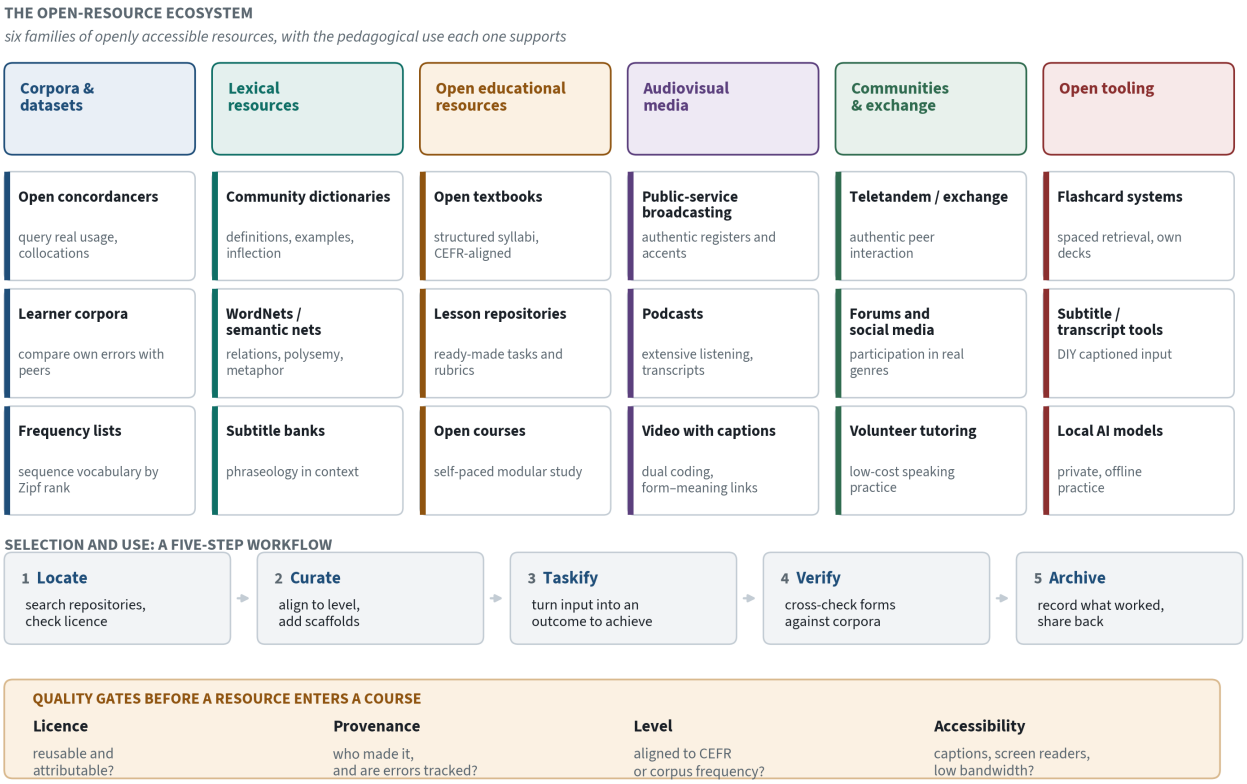


Figure 5. The open-resource ecosystem. Six families of openly accessible resources, a five-step workflow for turning resources into tasks, and the quality gates that should precede adoption.

7.2 Informal digital learning and the quality–quantity distinction

A large share of contemporary language learning happens outside any curriculum. Research on informal digital learning of English (IDLE) has produced one of the more practically useful findings in this area: the *diversity* of informal digital activities predicts vocabulary outcomes, whereas the *quantity* of time spent does not (Lee, 2019). Learners who combine form-focused and meaning-fo-

cused activities — watching, reading, chatting, gaming, making — do better than learners who spend equivalent time on a single activity. This result should temper the enthusiasm of platform dashboards that report streaks and minutes.

Studies of self-directed mobile learning similarly identify personalisation as the dimension learners exploit most, and authenticity and social connection as the dimensions they exploit least (Lai & Zheng, 2018) — an inversion of what designers assume, and a reminder that the affordances of a device are realised by learner practice rather than by its specification.

7.3 From resources to tasks

Abundance creates a curation problem that is pedagogically more demanding than scarcity. Figure 5 proposes a five-step workflow: locate, curate, taskify, verify, archive. The critical steps are the middle two. *Taskifying* converts a resource into a reason to use language: a podcast episode becomes a listening task with an outcome, not an item to be consumed. *Verifying* cross-checks the language in a resource against a corpus or dictionary, since open resources contain errors and since learners cannot always tell authentic irregularity from mistake.

Four quality gates should precede adoption: licence (reusable and attributable), provenance (who produced it, and are errors tracked), level (aligned to CEFR or corpus frequency), and accessibility (captions, screen-reader compatibility, low bandwidth). The last of these is not a minor consideration. A resource that assumes reliable broadband and a recent device excludes precisely the learners for whom open access matters most.

8. What the evidence actually shows

8.1 Reading effect sizes

Figure 6 assembles reported effect sizes from the syntheses discussed above. The bars are deliberately heterogeneous: they come from different literatures, different designs and different outcome measures, and the figure is intended to show the shape of the evidence rather than to rank the tools.

Four features of the pattern are worth naming. First, the largest values attach to interventions in which technology supplies something the learner would otherwise lack — large quantities of comprehensible input, immediate information about form, or unlimited low-stakes practice. Data-driven learning with corpora and captioned video both fit this description. Second, the smallest values attach to comparisons in which the technology is a different delivery channel for essentially the same instruction: the average advantage of computer-assisted instruction over face-to-face instruction is real but modest (d = 0.26; Plonsky & Ziegler, 2016), and synchronous computer-mediated interaction is statistically indistinguishable from face-to-face interaction (d = 0.13, not significant; Ziegler, 2016). That last finding should be read as good news rather than as a disappointment: it means that mediated interaction can substitute for face-to-face interaction without loss, which is precisely what learners who lack access to speakers need.

Third, the benchmarks matter. In second-language research, Cohen's conventional small/medium/large thresholds of 0.20/0.50/0.80 systematically underestimate the effects typically observed; field-specific benchmarks place small at approximately 0.40, medium at 0.70 and large at 1.00 (Plonsky & Oswald, 2014). A d of 0.60 that looks "medium" against Cohen's scale is close to "small" against the L2 scale. Fourth, almost none of these syntheses can say much about **retention** and **transfer**. Most

primary studies measure outcomes immediately or a few weeks after an intervention, on tasks closely resembling the treatment. The effect that matters — durable ability that generalises beyond the app — is the least measured.

Table 4. Evidence status by technology, with the characteristic limitation of the available research.

Technology	Strongest available evidence	Effect magnitude	Characteristic limitation
Corpora / data-driven learning	Meta-analysis of 64 studies	*d* = 0.95 (between groups); 1.50 (pre/post)	Predominantly advanced learners and academic writing
Captioned video	Meta-analysis of 18 studies	*g* = 0.99 listening; 0.87 vocabulary	Short treatments; productive outcomes weaker
Mobile apps (Duolingo, Babbel)	Independent efficacy studies	Receptive gains comparable to early university courses	Speaking develops less; interaction barely trained
Dialogue-based CALL / bots	Meta-analysis of 17 studies	*d* = 0.58, sustained on delayed tests	Small samples; short duration; varied outcome measures
Generative AI (writing)	Rapidly growing but mostly non-experimental evidence	Positive on immediate text quality	Few delayed or transfer measures; assessment validity
Generative AI (speaking)	Small controlled studies	*d* = 0.60 in one study with *n* = 60	Replication needed; novelty effects unresolved
VR / immersive environments	Systematic reviews	Positive but heterogeneous	Cost; novelty; small samples

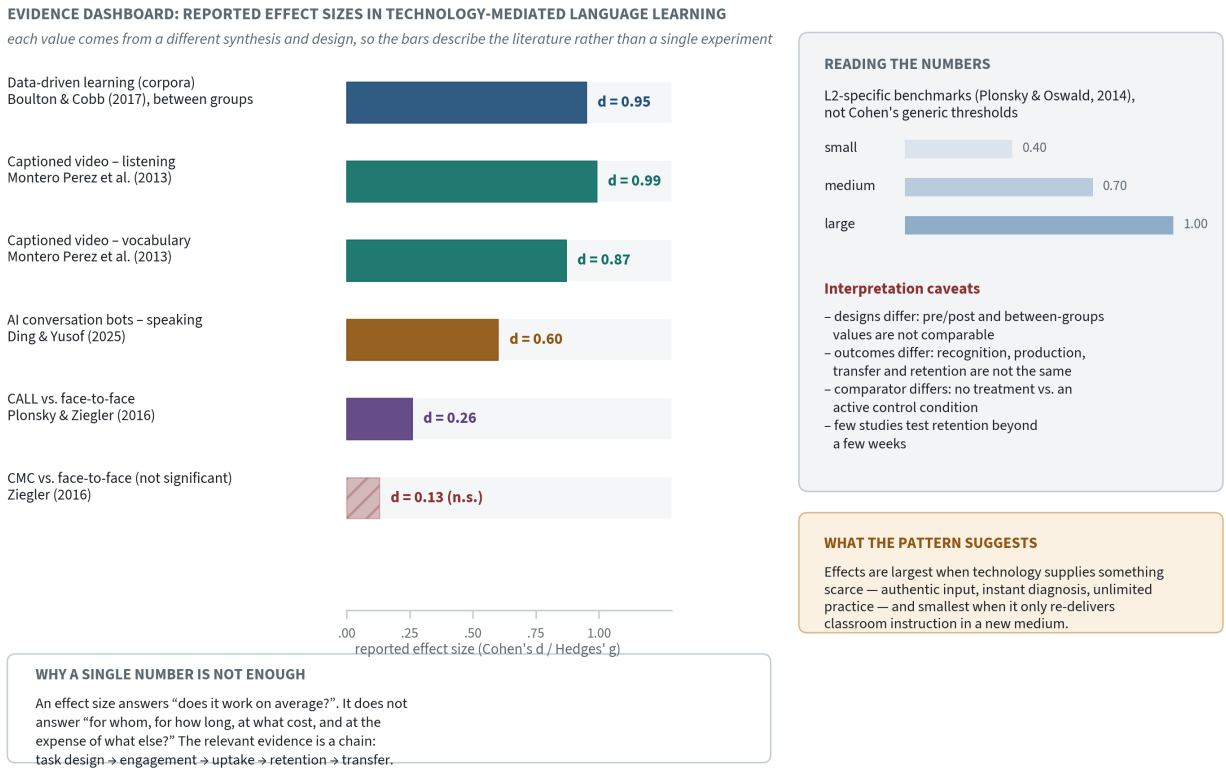


Figure 6. Reported effect sizes for technology-mediated language learning, placed against L2-specific benchmarks. Because designs, outcomes and comparison conditions differ across syntheses, the values are not directly comparable and should be read as an indication of magnitude rather than as a ranking.

8.2 Five methodological problems

The literature’s weaknesses are consistent enough to be listed, because they determine what can responsibly be claimed.

Novelty effects. New tools produce engagement because they are new. Studies conducted in the first months of a tool's availability cannot distinguish a learning effect from an interest effect, and follow-up periods are usually too short for the difference to emerge.

Absent or weak comparison conditions. A study comparing a chatbot with "no practice" demonstrates that practice helps. The informative comparison is with an active alternative: the same task with a human partner, or with a different technology.

Outcome measures that match the treatment. If the intervention trains discrete-item recognition and the test measures discrete-item recognition, a large effect is unsurprising and uninformative about communicative ability. The field needs measures of transfer — new tasks, new interlocutors, unassisted performance.

Self-report as outcome. Satisfaction, perceived usefulness and self-efficacy are legitimate constructs, and self-efficacy in particular is a genuine predictor of persistence (Bandura, 1997; Kittredge et al., 2025). They are not substitutes for measures of ability, and the frequency with which they are substituted is one reason the evidence base looks stronger than it is.

Supplier involvement and publication bias. A meaningful share of efficacy research on commercial products is conducted by, funded by, or reported by the companies concerned, and positive results are more likely to be published (Loewen et al., 2019). This does not make the findings false; it makes independent replication necessary.

8.3 What can be asserted with reasonable confidence

Taking the evidence as a whole, the following claims are defensible.

1. Technology-mediated language learning produces real learning gains, of a magnitude comparable to classroom instruction on the outcomes typically measured (Plonsky & Ziegler, 2016).
2. Specific technologies with a clear mechanism have larger effects: corpora for form-focused discovery, captions for listening and vocabulary, dialogue systems for speaking practice and anxiety reduction.
3. Bots and generative tools raise engagement, willingness to communicate and self-efficacy reliably; their effects on durable linguistic ability are less well established.
4. Effects depend more on task design, integration and learner variables than on tool identity.
5. The risk profile is dominated by validity and dependence rather than by technical failure.

9. A hypothetical roadmap

9.1 Purpose and status of the roadmap

The framework says what the components are; it does not say in what order to use them. Figure 7 therefore offers a hypothetical roadmap — a 26-week sequence from A1 to B2 for an adult learner studying independently with periodic human oversight. It is an illustration of sequencing logic, constructed from the mechanisms discussed above, not a validated curriculum, and the weekly hours are indicative rather than prescriptive.

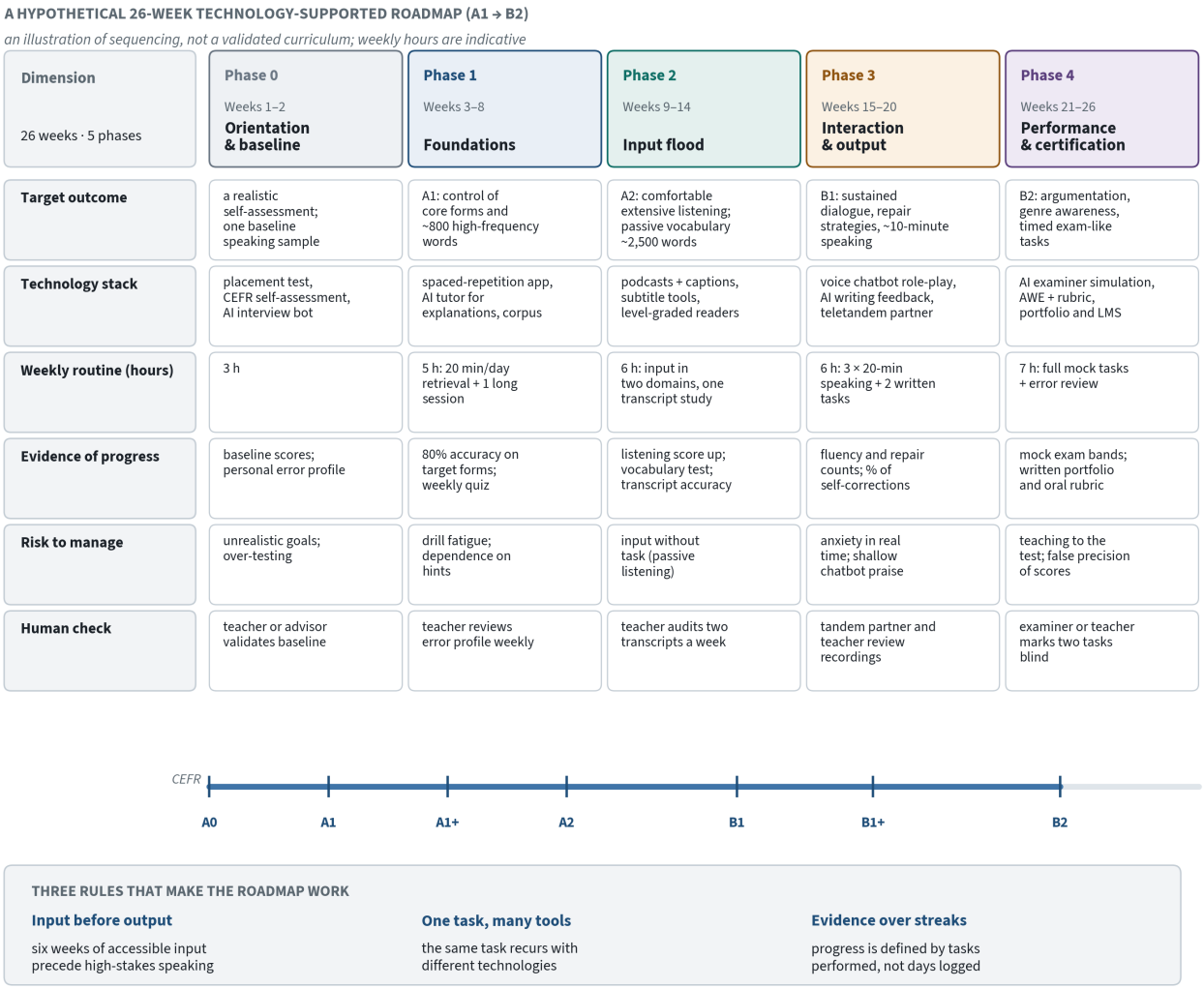


Figure 7. A hypothetical 26-week technology-supported roadmap (A1 → B2). Each phase specifies a target outcome, the technology stack, a weekly routine, evidence of progress, the principal risk, and the human check that closes the loop. The CEFR ladder below indicates the intended progression.

9.2 The logic of the sequence

Phase 0 (weeks 1–2): orientation. The purpose of the first fortnight is not to learn language but to establish a baseline and a map. A placement test, a realistic self-assessment and a recorded speaking sample produce an error profile that will be revisited throughout. The most common failure of self-directed study is not lack of tools but lack of a defined starting point.

Phase 1 (weeks 3–8): foundations. Early learning is dominated by the retrieval of high-frequency forms and vocabulary. This is the phase in which spaced-repetition systems and tutor-mode AI are most useful, because the content is discrete, the feedback is verifiable, and the risk of hallucination is manageable. The target is deliberately modest: control of core forms and roughly 800 high-frequency words.

Phase 2 (weeks 9–14): input flood. With a basic vocabulary in place, comprehension becomes possible and input becomes the most efficient use of time. Podcasts with transcripts, captioned video and level-graded reading dominate. The characteristic failure of this phase is passive consumption: listening that is never checked, vocabulary that is never recycled. Hence the requirement, in the roadmap, of a transcript study each week and a listening measure that is independent of the app.

Phase 3 (weeks 15–20): interaction and output. Only now does speaking become the centre of the routine, and only now is the learner able to sustain it. Voice-based role-play with an agent, AI-assisted writing with revision, and a human teletandem partner are combined. This ordering is deliberate: pushing output before there is language to push produces frustration, and it is the most common mistake in app-based study.

Phase 4 (weeks 21–26): performance and certification. The final phase reverses the logic of practice: instead of tasks designed to teach, the learner performs tasks designed to measure under realistic constraints — timed exam-like conditions, blind marking, a portfolio with revision history. This is also where the human check becomes non-negotiable, because it is the phase in which the learner is most vulnerable to false precision about their own ability.

9.3 Three rules that make the sequence work

Input before output. Six weeks of accessible input precede high-stakes speaking. The claim is not that output is unimportant but that it is expensive, and that its yield depends on the material available to work with.

One task, many tools. The same communicative task — describing a past weekend, complaining about a product, explaining a decision — recurs across phases with different technologies. This converts a collection of apps into a curriculum, and it makes progress visible in terms that are not specific to any one platform.

Evidence over streaks. Progress is defined by tasks performed and recorded, not by days logged. Streaks measure compliance with an application; portfolios measure ability. The distinction matters most for learners who are motivated by gamified systems and therefore most at risk of mistaking engagement for learning.

9.4 Adapting the roadmap

Three adaptations are worth noting. For **classroom-based learners**, the roadmap describes the out-of-class layer: the technology-mediated work that surrounds lessons, with the teacher supplying the human check. For **learners of less commonly taught languages**, the open-resource ecosystem is thinner and the input phase must be extended, while generative models are relatively more valuable for producing graded material — at the cost of a higher verification burden. For **older or less digitally confident learners**, the number of tools should be reduced rather than increased; a small number of well-understood tools used consistently outperforms a large number used intermittently, a finding consistent with research on self-directed mobile learning (Lai & Zheng, 2018).

10. Risk, ethics and equity

10.1 The accuracy problem

Generative models produce fluent, plausible and sometimes false language. For language learners the problem is sharper than for other users, because the learner's ability to detect falsity is precisely what is being developed. A fabricated collocation presented confidently is not merely an inconvenience; it is input, and it will be acquired.

Three mitigations are available and should be standard. **Retrieval grounding** — constraining generation by reference to a corpus or dictionary — converts an oracle into a checkable source. **Explicit verification routines** — teaching learners to check an unfamiliar collocation in a corpus or a learner's dictionary before adopting it — turn a limitation into a skill, which is itself a metacognitive outcome. **Attribution and escalation** — systems that indicate their confidence and route uncertain cases to a human — preserve trust where it is warranted.

10.2 The dependence problem

Cognitive offloading is rational. If a tool answers in three seconds, the effortful retrieval that produces durable memory seems pointless, and learners will rationally decline to make it (Risko & Gilbert, 2016). The result is a gap between *performance with the tool* and *ability without it*, and that gap is invisible to the telemetry that platforms collect.

The pedagogical response is to design tasks in which the tool is deliberately unavailable at the moment of performance: unaided retrieval, closed-book production, live conversation with a human. This is not Luddism; it is the same principle that makes spaced retrieval effective. A useful working rule is that the tool should be present during *preparation* and absent during *performance*, at least some of the time.

10.3 The bias problem

Speech recognition and automated scoring are trained on distributions of speech and writing that are not neutral with respect to accent, dialect, first language, gender, age or disability. Where a model has seen less of a variety, it errs more, and the error is systematically attributed to the learner rather than to the system. Three consequences follow. Assessment scores may be unfair in ways that are invisible to the institution. Learners may be pushed towards a prestige accent that is neither necessary for intelligibility nor appropriate as a goal. And languages and varieties with less training data receive inferior tools, reproducing existing hierarchies of attention.

Institutions adopting AI-based assessment should therefore require, at minimum, evidence of performance disaggregated by first language and accent group, a documented process for appeal, and a human rater for consequential decisions.

10.4 Privacy, surveillance and the datafication of learning

Language-learning applications collect unusually intimate data: voice recordings, written production, error histories, and in the case of speaking partners, conversational content. This data is used to improve models, to personalise content and to feed engagement analytics, and its retention is often poorly specified. The ethical issues are not hypothetical — they concern minors, biometric data and inferences about proficiency that may follow a learner across institutions (Hockly, 2023; Williamson & Eynon, 2020).

Four practical requirements follow: data minimisation, so that only what is needed for the pedagogical function is collected; local processing where feasible, which is now realistic with small models running on learner devices; retention limits with clear deletion guarantees; and transparency about what the system infers and how it is used.

10.5 Equity and the amplification of advantage

Digital tools do not distribute themselves evenly. Learners with reliable connectivity, newer devices, quiet space and the digital literacy to navigate multiple platforms benefit disproportionately; learners without these things may be worse off than in a well-resourced classroom, because the alternative to a well-used tool is not always a well-taught lesson. The same logic applies to AI: a model that requires a paid subscription and English-language prompting skill advantages precisely those who already have advantages.

Reviews of AI in education have repeatedly noted that questions of equity, inclusion and the political economy of educational technology are under-represented in the research literature (Zawacki-Richter et al., 2019; Holmes & Tuomi, 2022). The practical responses are unglamorous: offline-capable resources, low-bandwidth versions, institutionally licensed rather than individually purchased tools, training in the local language, and evaluation that disaggregates outcomes by access conditions rather than reporting a single average.

10.6 Academic integrity

The integrity crisis in language assessment is real but has been misdiagnosed. Detection software does not work reliably, and treating every learner as a suspect damages the relationship on which feedback depends. The durable solutions are assessment designs that make outsourcing pointless: in-class writing with a revision trail, oral defence of written work, tasks that require reference to locally provided materials, and process portfolios. Generative models are, in this respect, less a threat to assessment than a prompt to assess what was always worth assessing.

11. Implications and research agenda

11.1 For learners

1. Choose tools by the mechanism they activate, not by their feature list: ask which of input, interaction, feedback and autonomy a tool actually supplies.
2. Use generative models for preparation and reflection, and perform without them at regular intervals.
3. Diversify informal activity. Variety of activity predicts vocabulary outcomes better than volume of time (Lee, 2019).
4. Treat scores from apps as formative signals, not as measurements of ability.
5. Verify unfamiliar language against a corpus or a dictionary before adopting it.

11.2 For teachers

1. Design the task first, then choose the technology; a mediocre tool in a good task outperforms an excellent tool in no task.
2. Automate diagnosis and variation, retain judgement: the division of labour in Figure 4 is a reasonable default.
3. Require uptake. Feedback that does not produce a subsequent action is decoration.
4. Audit machine feedback periodically against your own judgement, and keep a record of disagreements — the disagreements are the curriculum.

5. Teach critical AI literacy explicitly, including verification routines and the limits of automated scoring (Godwin-Jones, 2024).

11.3 For institutions

1. Procure on evidence, not on demonstration: require validation against human raters and disaggregated fairness data.
2. Protect the human check as a structural feature of any AI-mediated course, with time allocated to it.
3. Redesign assessment around what cannot be outsourced.
4. Negotiate data terms explicitly: retention, secondary use, portability and deletion.
5. Fund the open layer. Institutional support for corpora, OER and repositories is cheap relative to licences and produces public goods.

11.4 Research agenda

The field does not primarily need more studies of whether a tool is liked. It needs the following:

- **Delayed and transfer designs.** Outcomes measured weeks or months after the intervention, on tasks and with interlocutors not used in the treatment.
- **Active comparators.** The same task with a human, with a different technology, and with no tool, so that the contribution of the tool can be isolated.
- **Disaggregated reporting.** Effects by proficiency, age, first language, accent, device and connectivity, rather than a single average.
- **Process data linked to outcomes.** Transcripts, revision histories and interaction logs analysed alongside learning measures, rather than engagement metrics alone.
- **Validity studies of automated scoring.** Systematic comparison with trained raters across varieties and task types, published whether or not the results are favourable.
- **Independent evaluation of commercial products.** Commissioned studies are a starting point, not a substitute.
- **Studies of teacher practice.** What teachers actually do with these tools, and what support they need, remains under-researched relative to studies of learner use.

12. Conclusion

The question posed at the outset was not whether digital technology helps people learn languages — it plainly can — but under what conditions it does so, and when it merely produces the sensation of progress. The answer defended here is that the effects are conditional on mechanism. Technology improves language learning when it supplies comprehensible input in quantity, when it creates occasions for negotiated interaction and pushed output, when it delivers feedback that is timely, specific and acted upon, and when it supports autonomy without removing the productive difficulty that learning requires. It does much less when it re-delivers conventional instruction through a new medium, when it substitutes praise for repair, and when it measures engagement instead of ability.

The present moment is unusual in that the supply of the first three of these has increased enormously and cheaply, while the fourth — autonomy with productive difficulty — has become harder to protect. Generative models are extraordinarily good at removing friction, and friction is where a good deal of learning lives. The most important design decision in contemporary language education is therefore not which model to adopt but where to keep the difficulty: which errors should be corrected, which tasks should be performed unaided, and which judgements should remain human.

The framework, taxonomy, workflow and roadmap offered here are tools for making those decisions explicit. They are not a substitute for evidence, and the evidence base is thinner than the volume of publication suggests. But the direction of the argument is clear, and it is a hopeful one: the resources now available to a motivated learner of any language are unprecedented, and the pedagogies required to use them well are not new. They are the ones the field has been describing for forty years — comprehensible input, meaningful interaction, contingent feedback, and the gradual transfer of responsibility to the learner. What is new is that these can now be supplied at scale, at any hour, to almost anyone. Whether they are supplied well is a design question, and it remains ours.

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Appendix A. Figure and table inventory

Figures. Figure 1 — integrative framework: three technological layers, four pedagogical mechanisms, four outcome families (Section 3). Figure 2 — architecture of an AI-in-the-loop environment (Section 4.3). Figure 3 — taxonomy of AI conversational agents by pedagogical role (Section 5.1). Figure 4 — the automated feedback loop and the division of labour (Section 6.1). Figure 5 — the open-resource ecosystem, workflow and quality gates (Section 7.1). Figure 6 — evidence dashboard: reported effect sizes against L2 benchmarks (Section 8.1). Figure 7 — hypothetical 26-week technology-supported roadmap (Section 9).

Tables. Table 1 — three generations of technology in language learning, by metaphor, mechanism and weakness (Section 2.1). Table 2 — four modes of generative-AI use, with the mechanism each activates and its principal risk (Section 4.1). Table 3 — design parameters for a task-based bot session (Section 5.3). Table 4 — evidence status by technology, with the characteristic limitation of the available research (Section 8.1).

Appendix B. A prompt specification for a task-based bot

The following specification is offered as an illustration of the design principles in Section 5.3. It is written so that a teacher can adapt it to any language and level without technical knowledge.

> **Role.** You are a hotel receptionist in [city]. You speak only [target language].
>
> **Situation.** The learner has a confirmed booking, but the room is not available. You may offer alternatives, but only if the learner asks for them.
>
> **Level.** Speak at CEFR [A2/B1/B2]. Use short sentences. Introduce no more than two words the learner is unlikely to know in any turn, and gloss them briefly in the target language.
>
> **Do not** solve the problem until the learner has (a) stated the problem, (b) requested a solution, and (c) either accepted or rejected the offer with a reason.
>
> **Correction policy.** After each learner turn, recast at most one error implicitly by using the correct form in your reply. Do not comment on errors explicitly and do not list corrections.
>
> **Repair.** If the learner's message is unclear, ask one clarification question rather than guessing.
>
> **Closure.** When the situation is resolved, end the role-play and produce, in the target language, a three-line summary of what was agreed. Then ask the learner to write the same summary without looking at yours.
>
> **Escalation.** If the learner asks a question about grammar, answer it in one sentence and return to the role-play.

The specification illustrates three points made above: that the pedagogical work happens in the constraint rather than the model; that error policy is a design decision with consequences for fluency; and that closure is what converts a conversation into evidence.